

MISY350: Business Application Development II

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Course Description

This course will focus on current technologies and tools that are used to develop web applications in a business environment. We will cover a wide range of client-side and server-side web development technologies, including HTML, CSS, JavaScript and jQuery, Bootstrap, SQL, and Python Flask. In addition, we will learn several web-based data analysis technologies such as visual analytics (FusionCharts.js). Preliminary knowledge about relational databases is necessary but not required for you to work on database-related projects. Upon the completion of this course, you will be able to develop a full-stack web application in a business setting.

Course Materials

Recommended Textbooks:

"Internet & World Wide Web: How to Program", 5th edition, Deitel et al., ISBN: 0-13-978-0132151009.

"Flask web development: Developing web applications with Python", 2nd edition, M. Grinberg, ISBN: 013-978-1491991732.

Hardware: A Windows or Mac computer is required. Tablets or Chromebooks are not acceptable.

Software: [Visual Studio Code Editor](#); Docker Desktop

Other course materials: Available on Canvas.

Office Hours and Email Policies

The instructor will hold in-person office hours at the times announced in class. You are also encouraged to ask course-related questions and help each other at the Canvas discussion forum. Please make sure to check existing postings in the discussion forum before you post your question. It is very likely that your question has been asked and answered in the forum.

I will NOT answer coding questions through emails. You MUST make every effort to come to my office hours or to utilize the discussion forum for homework-related questions. In case you need to email me for an issue that cannot be shared in the public forum, the following email conventions must be followed. If they are not I may not receive or answer your email. To ensure that your email is appropriately filtered, you MUST put "MISY350" in the subject line. All emails must include a greeting (example: "Hi Dr. Wang"), a subject (the topic of your email), and your full name (example: "Bob Jones").

Grading

Assignments	20%
In-Class Quizzes	15%
Online Discussions	10%
Midterm Exams	25%
Team Project	30%

Total	100%

The grade distribution will NOT be curved. Letter grades are assigned as follows:

93% and above	A
90-92%	A-
87-89%	B+
83-86%	B
80-82%	B-
77-79%	C+
73-76%	C
70-72%	C-
67-69%	D+
63-66%	D
60% to 62%	D-
Below 60%	F

Integrity and Honor Code Policy

The Academic Honesty Policy at the University of Delaware states "Students must be honest and forthright in their academic studies." Students are expected to do their own work and neither give nor receive unauthorized assistance." The standards of behavior and expectation of honesty apply in all academic settings, including in-person, online and hybrid. If there are questions regarding course expectations, please contact your professor before beginning the work. Canvas has many methods of tracking your use and access while using the technology. Any potential instances of dishonesty may be forwarded to Community Standards & Conflict Resolution for action.

Assignments

All homework assignments must be INDIVIDUALLY completed. You can ask for help from other students. But you must be able to independently complete the homework. Copied code from any source is considered as an Honor Code violation and will be reported to the Honor Court. Digital copies of assignments are distributed from and submitted to the Canvas course site.

A late submission will receive a 10% penalty if submitted less than 24 hours late. Late submissions will NOT be accepted if submitted more than 24 hours past due. No appeal to the late penalty will be accepted if your own technical problems prevent you from submitting homework on time. Your own technical problems include computer hardware and software issues, disconnected Internet, power outage (unless the entire town of Blacksburg goes out), car problems, etc. It is generally not a good idea to submit your homework within the last hour before the deadline. Please get it done early! You MUST leave enough time to complete the homework submission process and to receive a confirmation message from Canvas after submission. If you did not receive the confirmation from Canvas, you should check the submission status and resubmit it. It takes time to upload files. It is your own responsibility to make sure that your files have been submitted successfully.

Assignment grades can only be **appealed within 3 calendar days** after the grades are released. Absolutely no appeal will be accepted after that appeal period.

In-Class Quizzes

At the completion of each lecture, you are required to complete an online quiz for an evaluation during class time. The purpose is to make sure that you studied the lecture carefully and learned from it. The quizzes are open-book and open-notes. You must be physically present to take each in-class quiz. No make-up quiz will be allowed for missing quizzes even for approved/documented excuses. Class attendance and participation are key to your success in this class as many homework assignments will be completed during class time. In that regard, the in-class quizzes also serve as the attendance check.

The lowest quiz grade will be dropped at the end of the semester.

Discussion Participation

You are required to participate in the online discussion related to each lecture on Canvas. You can participate the discussion before and/or during class time for which the discussion topic is the subject of the class. Discussion participation is a graded activity.

You are required to contribute at least 2 postings in each discussion thread, which can be a combination of question postings, answering postings, and/or knowledge sharing postings. The postings must be relevant to the teaching material for the day in order to receive credit.

You can ask homework related questions in the question posts. It is fine to share a small segment of code directly related to the problem discussed. But copying the entire solution is prohibited and will be considered as an Honor Code violation.

In the knowledge sharing posts, you can talk about anything that you think is useful, including but not limited to: relevant knowledge not covered in the lecture, online articles and sources, alternative solutions to the lecture demos and exercises, tips, and tricks.

Exams

There are three midterm tests. The first two are closed-book, closed-notes, and non-cumulative exams. The last one is an open-book and open-notes one.

You must take the tests at the designated class times. If you miss an exam for any reason except serious and/or debilitating illness/injury, you will receive a ZERO grade for that test. If you cannot take an exam at the designated class time, please let me know before the exam date.

There is no final comprehensive exam, which has been replaced by a team project, the details of which will be provided later.

Team Project

This team project will help you gain experience on building a Web application, including use case analysis, design, implementation, and testing. This experience will not only improve your understanding of our course material, but also benefit you in the long run regarding your future employment opportunities. Details of the team project will be released later.

The project grade will be given based on the following deliverables:

Midterm progress report 10%
Project Presentation Slam 10%
Final project submission 10%

Get Ready

1. Clean up your computer, check and install system updates, do not upgrade your computer to a major operating system release (e.g., upgrade from macOS Ventura to macOS Sonoma) until the semester is over.
2. Download and install [Visual Studio Code](#).
3. Get the recommended textbook from the bookstores or other sources (however, lectures and homework assignments are not from the textbook) if you prefer to keep a reference book handy.
4. Review the course curriculum on the Module page.

(updated on March 8, 2025)

DATE	LECTURE	Chapter	Type	ASSIGNMENT DUE
05-21	1 Course Syllabus and Introduction		Lecture	
	2 Intro. to Internet and Web Applications	D1	Lecture	
05-23	3 HTML5 I	D3	Lab	
	4 HTML5 II	D4	Lab	
05-26	5 CSS I	D5	Lab	
	6 CSS II	D5	Lab	HW1
05-28	7 JavaScript I: Basics " What can you do with JavaScript " Team Project Requirements	D6-8 handouts	Lab	
	8 JavaScript II: Functions and Arrays	D9-10	Lab	HW2
05-30	9 Test 1 (HTML5, CSS)			
	10 Software Development Life Cycle (SDLC)			
06-02	11 JavaScript III: Objects, DOM, and Events	D11-13	Lab	
	12 JavaScript III: Objects, DOM, and Events	D11-13	Lab	HW3
06-04	13 Responsive Design Framework: Bootstrap	handouts	Lab	
	14 Bootstrap Studio	handouts	Lab	HW4
06-06	15 Project Workshop			HW5
	16 Project Workshop			Project Midterm Progress Report (6/1)
06-09	17 Test 2 (JavaScript, BOOTSTRAP)			
	18 Python I: Intro to Programming with Python and Flask Overview of database concepts & MySql	handouts	Lab	
06-11	19 Python II: Forms processing	D19	Lab	
	20 Python II: Forms processing	D19	Lab	HW6
06-13	21 Python III: State Management and Ajax Applications	handouts	Lab	

	22	Python III: State Management and Ajax Applications	handouts	Lab	HW7	
06-16	23	Visual Analytics	handouts	Lab		
	24	Project Workshop			HW8	
06-18	25	Project Workshop				
	26	Exam 3				
06-20	27	Team Project Presentation				
	28	Team Project Presentation			Project Final Deliverable	